Molecular photoionization cross sections by the Lobatto technique. I. Valence photoionization

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A method for the calculation of electronic continuum wave functions is presented which is based on the logarithmic derivative version of the Kohn (LDK) variational principle. The variational principle is cast into algebraic form by introducing a finite basis set that consists of spherical harmonic Gaussian-type functions (GTOs) and of Lobatto shape functions with the latter representing the translational part of the basis. A local effective potential which is obtained from density functional theory results in fairly accurate photoionization cross sections. Also studied are asymptotic corrections to the effective potential for the photoelectron which, in many cases, lead to improved results. The Lobatto procedure is applied to the diatomics N_2 and CO and to benzene which may be regarded as a prototype for larger non spherical symmetric systems for which the method is targeted. For the two diatomics, results in excellent agreement with experiment have been found. For benzene the results are compared to those obtained by the Stieltjes–Tchebychev (ST) imaging technique and by the continuum multiple scattering (CMS) method which both have been applied to similar effective local potentials. Comparison with the ST imaging technique shows that the LDK Lobatto (LDKL) method provides qualitatively similar results, but the LDKL cross sections are of higher resolution and allow a more detailed analysis because of the explicit determination of the continuum wave function. For most of the valence orbitals of benzene the CMS method does not lead to satisfactory agreement with experiment due to the well-known deficiencies of this technique. The LDKL method implemented with a combined basis set does not suffer from the limitations of the ST and the CMS methods, but remains applicable to larger-size molecules.

I. INTRODUCTION

The interpretation of molecular photoelectron spectra benefits considerably from calculations of the photoelectron dynamics which can be characterized by the differential cross section for the ionization as a function of photon energy.[1] The calculation of cross sections often is based on a one-electron picture, in which the "active" electron is excited out of the initial molecular orbital into a final unbound state through the absorption of the incident photon. The remaining electrons are assumed to be unaffected by the ionization process (frozen orbital approximation) or relaxation processes are taken into account by modifications of the "final state" potential, e.g., by the use of a transition state potential.[2,3] Obviously, a one-electron framework prevents the description of many-electron features like autoionization resonances. Usually photoelectron spectra exhibit many structures which can be explained satisfactorily within a one-electron approach. Furthermore, an accurate one-electron calculation may allow the separation of one-electron and many-electron features in a given experimental spectrum.

At present, there are four major types of techniques in use for the determination of electronic continuum wave functions or for the direct calculation of molecular photoionization cross sections, respectively. Firstly, there are the Stieltjes–Tchebychev (ST) imaging methods[4,5] that avoid the explicit determination of the continuum wave function by deriving an approximate cross section from discrete oscillator strengths associated with virtual levels which are obtained from a bound state calculation. This approach involves only conventional L^2-basis functions and has been successfully applied at the basis of Hartree–Fock and of configuration interaction (CI) calculations to many systems, for example, CO, CO_2, H_2CO,[6] or H_2O.[7] ST imaging results based on a local potential from LDF theory[8] will be compared to the results in this work. Major disadvantages of ST imaging procedures are the rather limited energy resolution which is caused by inherent numerical instabilities and the lack of information on the angular distribution of the photoelectrons since the explicit calculation of the continuum wave function is circumvented. Second, there are methods which expand the electronic continuum wave function into a finite basis set over the entire space, e.g., the iterative Schwinger method[9] and the complex Kohn method.[10] The third type of method takes advantage of the fact that it is sufficient to describe the interactions between the photoelectron and the remaining ion with high accuracy only within a finite region (determined approximately by the initial state wave function) while assuming a simple asymptotic form for the unbound photoelectron state in the remaining space. Methods involving this spatial partitioning are the widely used R-matrix method[11] and the finite volume variational method (FVVM).[12] The logarithmic derivative Kohn (LDK) variational procedure[13,14] also employs a spatial partitioning; the application of this method to the determination of electronic continuum wave functions[15,16] will be elaborated in this work.

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All these scattering methods exhibit similarities in the calculational procedure as they are implemented in algebraic form. Due to the discretization in the ansatz space, matrix elements for a pertinent set of operators are evaluated and the (variational) problem is represented by a set of linear equations.

A fourth and older, but well-established nonalgebraic method for the calculation of continuum wave functions is the continuum multiple scattering (CMS) method which draws its efficiency from a muffin-tin partitioning of the effective molecular potential.[2,17,18] Though there are well-known deficiencies connected with this shape approximation to the potential,[19] the method is still in use.[20,21]

Of all the methods mentioned so far, only the ST imaging approach and the CMS model with their well-known drawbacks have been applied to larger systems, e.g., larger than formaldehyde, H_2CO. Here we present a method that allows the treatment of larger systems, but is free of the deficiencies of the former procedures. It employs a local effective potential obtained from density functional theory which has been shown to lead to a fairly good description of the photoionization process using a ST technique.[8] The explicit calculation of continuum wave functions is based on the LDK variational principle[13,14] implemented for a finite basis set expansion. This basis set is chosen as a combined set of Lobatto shape functions and of conventional Gaussian-type functions (GTO). The former lead to a highly flexible representation of the long-range part of the wave function and allow a very efficient calculation of the occurring integrals. The latter are ideally suited to expand the short-range part of the continuum wave function influenced by the singularities of the Coulomb potential. Moreover, the initial state calculation is done using the same set of GTOs, therefore resulting in a convenient procedure for the evaluation of a large part of the matrix elements.

II. METHOD

A. The finite volume variational principle

The Kohn variational principle originally was devised for the application to one-dimensional problems.[13] Recently an extension to three-dimensional scattering has been proposed and successfully applied[14,22,23] to molecular scattering, i.e., in the area of reaction dynamics. The closely related FVV method, designed for the calculation of photoionization cross sections, has been applied to H_2^+.[12] The relationship between the FVV method and the present LDK approach is discussed in the Appendix. Because our utilization of the LDK variational principle in the context of a general one-electron problem has been described in detail elsewhere[15,16] only a short review of the basic formalism is given in the following. Then, we turn to specific problems arising from the choice of the local potential which underlies the calculation of molecular photoionization cross sections in many-electron systems.

The logarithmic derivative version of the Kohn variational principle[14,22–24] at a given energy E for a potential V in a finite Volume $\mathscr{V}$ starts from the following functional:

$$\mathscr{L}[\varphi,\psi]:=\langle\nabla\varphi|\nabla\psi\rangle_{\mathscr{V}}+2\langle\varphi|V-E|\psi\rangle_{\mathscr{V}}=\langle\varphi|\partial_n\psi\rangle_{\partial\mathscr{V}} \tag{1}$$

with the second equation being valid at stationarity of $\mathscr{L}[\varphi,\psi]$ with respect to the variation of φ.

The indices $\mathscr{V}$, $\partial\mathscr{V}$ indicate integration over an arbitrary finite volume and the corresponding surface integral, respectively. The expression at stationarity of the functional $\mathscr{L}$ with respect to variation of φ may be obtained by applying Greens theorem and by introducing a constraint which forces the variation of φ to vanish at the surface $\partial\mathscr{V}$ of the chosen volume $\mathscr{V}$.[14,16] The stationary condition $\delta\mathscr{L}=0$ then leads to a solution ψ of the Schrödinger equation at the given energy E within that volume. Choosing the finite volume as a sphere of radius S allows one to take advantage of the partial wave expansion of the wave function into (real) spherical harmonics,

$$\psi(r,\Omega)=\frac{1}{r}\sum_k R_k(r)Y_k(\Omega), \tag{2}$$

valid within the volume $\mathscr{V}$. In Eq. (2) the compound index k denotes the angular momentum quantum numbers (l,μ) of the spherical harmonics $Y_k(\Omega)$, whereas $R_k(r)$ is the corresponding radial function. As is typical for a continuum problem, a set of linearly independent degenerate solutions ψ exists. If one represents each solution of the form (2) by a column of the matrix $\mathbf{R}$ and collects the spherical harmonics in a row vector $\mathbf{Y}$ one arrives at the more compact notation

$$\Psi^{\text{in}}=\frac{1}{r}\mathbf{YR}. \tag{3}$$

The volume $\mathscr{V}$ is chosen large enough so that all short-range components of the potential V, i.e., all higher multipole moments, have decayed and the potential relevant for photoionization can be assumed to be purely Coulombic in the external region. We then may write the solutions outside $\mathscr{V}$ as linear combinations of Coulomb waves,

$$\Psi^{\text{out}}=\frac{\mathscr{N}}{r}\mathbf{Y}(\mathbf{FA}+\mathbf{GB}). \tag{4}$$

In Eq. (4) $\mathbf{F}$ and $\mathbf{G}$ are diagonal matrices built of the familiar regular and irregular Coulomb waves, respectively.[25] These Coulomb waves are adapted to the charge of the remaining system (see Sec. III B). The factor $\mathscr{N}=(\pi^2E/2)^{-1/4}$ ensures the proper normalization to the energy scale. The real matrices $\mathbf{A}$ and $\mathbf{B}$ are determined by matching the solutions $\mathbf{R}(r)$ and their derivatives across the surface $\partial\mathscr{V}$. All scattering information, including the S-matrix,[12] may be expressed through $\mathbf{A}$ and $\mathbf{B}$. For the description of a photoionization process, boundary conditions corresponding to incoming spherical waves are required,[12,26]

$$\Psi^{(-)}=\Psi^{\text{out}}(\mathbf{A}+i\mathbf{B})^{-1}. \tag{5}$$

The total photoionization cross section corresponding to a given final energy E (in a.u.) of the photoelectron is then obtained as

$$\sigma_{\text{tot}}=\frac{4}{3}\pi^2\alpha a_0^2(I+E)\mathbf{P}^T(\mathbf{A}^T\mathbf{A}+\mathbf{B}^T\mathbf{B})^{-1}\mathbf{P}, \tag{6}$$

where a_0 is the Bohr radius, α the fine structure constant, and I the ionization potential of the initial state in a.u. The column vector $\mathbf{P}$ of the transition moments (in a.u.) is given by

$$\mathbf{P}=\langle\Psi^{\text{in}}|\mathbf{r}|\psi_0\rangle_{\mathscr{V}}, \tag{7}$$

where one assumes that the initial state ψ_0 is confined to the interior of the volume $\mathscr{V}$. In the following applications we will use experimental values for the ionization potentials. Of course, in order to obtain purely theoretical results without resorting to experimental data, ionization potentials could also be determined by appropriate calculations, e.g., by transition state calculations or by ΔSCF calculations. The total photoionization cross sections will be quoted in units of megabarns ($1\ \text{Mb}=10^{-22}\ \text{m}^2$).

B. Algebraic form of the variational principle, introduction of basis functions

The variational problem is cast into algebraic form by introducing a finite basis which is flexible enough to represent molecular continuum states inside the volume $\mathscr{V}$, yet allows a rapid evaluation of the resulting matrix elements. In keeping with the partial wave expansion of Eqs. (2) and (3), we start with a basis Ξ of order $M+1$ of one-center functions,

$$\chi_{ik}(r,\Omega)=\frac{u_i(r)}{r}\,Y_k(\Omega). \tag{8}$$

The chosen radial functions $u_i(r)$ are Lobatto shape functions[14] (see Sec. II C) which satisfy the LDK boundary conditions,[13,14]

$$u_i(S)=\delta_{i0},\quad u_i(0)=0,\quad (i=0,...,M). \tag{9}$$

In order to take the multicenter nature of the wave function into account we augment the basis Ξ by GTOs which are, as common in electronic structure calculations, centered at the nuclei.[27] These basis functions will be referred to as "bound"-type basis functions whereas the Lobatto shape functions will be referred to as "free"-type basis functions. The volume $\mathscr{V}$ has to be chosen large enough such that the "bound"-type functions fulfill the LDK boundary condition (9), i.e., are zero on the surface $\partial\mathscr{V}$ to acceptable accuracy. To simplify the notation of the following derivation, we will not refer to these additional "bound"-type basis functions explicitly. The set of degenerate scattering functions is now specified by requiring each function to be proportional to a given spherical harmonic Y_k at the surface $\partial\mathscr{V}$,

$$\psi_k=\chi_{0k}+\sum_{k'}\sum_{i=1}^{M}\chi_{ik'}c_{ik'}^{k}. \tag{10}$$

In this expansion this is assured by appropriately fixing the coefficient of the basis function χ_{0k} which does not vanish at the surface. Furthermore, one exploits the fact that the "bound"-type basis functions contributing to ψ_k within $\mathscr{V}$ are zero at the surface $\partial\mathscr{V}$. Obviously, these functions are linearly independent on the surface $\partial\mathscr{V}$ since $\psi_k(S,\Omega)=S^{-1}Y_k(\Omega)$. The expansions (10) read in matrix form

$$\Psi^{\text{in}}=\Xi\tilde{\mathbf{C}}\ \text{with}\ \tilde{\mathbf{C}}=\begin{pmatrix}\mathbf{E}\\ \mathbf{C}\end{pmatrix}. \tag{11}$$

In Eq. (11) the columns of the matrix $\tilde{\mathbf{C}}$ represent the set of linearly independent solutions ψ_k. The row vector Ξ consists of the "free"- and the "bound"-type basis functions. The partitioning of the coefficient matrix $\tilde{\mathbf{C}}$ as given in the second part of Eq. (11) implies an ordering of the basis functions where the "surface" functions χ_{0k} are placed first. The appearance of the unit matrix $\mathbf{E}$ in the coefficient matrix $\tilde{\mathbf{C}}$ guarantees that the scattering functions ψ_k have a fixed value on the surface $\partial\mathscr{V}$ as required in the LDK variational principle.[13]

The matrix elements required for building up the functional $\mathscr{L}$ are given by

$$M_{ik,i'k'}=\langle\nabla\chi_{ik}|\nabla\chi_{i'k'}\rangle_{\mathscr{V}}+2\langle\chi_{ik}|V-E|\chi_{i'k'}\rangle_{\mathscr{V}}. \tag{12}$$

The stationarity of the functional with respect to the coefficients $\mathbf{C}$ leads to a system of linear equations,

$$\mathbf{M}_{11}\mathbf{C}+\mathbf{M}_{10}=\mathbf{0}. \tag{13}$$

The value of $\mathscr{L}$ at stationarity,

$$\mathbf{L}=\mathbf{M}_{00}+\mathbf{M}_{01}\mathbf{C}, \tag{14}$$

represents the generalized logarithmic derivatives in Eq. (1). In Eqs. (13) and (14) the real matrices $\mathbf{M}_{11}$, $\mathbf{M}_{10}$, $\mathbf{M}_{01}=\mathbf{M}_{10}^T$, and $\mathbf{M}_{00}$ consist of matrix elements of the energy-dependent functional; the partitioning of the matrix $\mathbf{M}$ is induced by that of the basis functions according to their "surface" behavior [see Eq. (11)]. Equations (10)–(13) may be generalized in a straightforward manner to include the "bound"-type functions. We will refer to this procedure as the "LDKL method" even when applied with a combined basis set of Lobatto shape and GTO functions.

The calculation of the coefficients $\mathbf{C}$ in Eq. (13) formally involves the inversion of the energy-dependent matrix $\mathbf{M}_{11}$. In general, this matrix is of rank N, where N is the number of "volume" basis functions, and thus invertible. If the energy E under consideration is very close ($\lesssim 10^{-5}$ eV) to an eigenvalue of the Hamiltonian operator projected onto the "volume" basis functions, the matrix $\mathbf{M}_{11}$ will no longer be invertible. This phenomenon is related to the "spurious resonances" found in other scattering variational principles (for a detailed discussion see Ref. 28). However, we have found these singularities never to present a problem in any of our calculations.

C. Specification of the radial basis, calculation of matrix elements

The choice of the radial basis $u_i(r)$ as Lobatto shape functions results in a highly efficient computational procedure for the various matrix elements.[14–16] The Lobatto shape functions are the Lagrange interpolation polynomials (of order $M+1$) corresponding to the set of $M+2$ nodes $\{r_j\}$ and weights $\{\omega_j\}$ associated with a Gauss–Lobatto quadrature.[29] The latter is a Gaussian integration scheme with the end points of the interval being the preassigned abscissas $r_0=S$ and $r_{M+1}=0$. The Lobatto shape functions $u_i(r)$ are localized near the nodes r_i and obey the following "orthogonality" condition:[14]

$$u_i(r_j)=\delta_{ij}, \tag{15}$$

which simplifies the integral evaluation significantly. For example the value of integrals containing a Lobatto shape function $u_i(r)$ only depends on the value of the integrand at a single radial point, namely at r_i. Thus, no further summation with respect to the radial nodes is necessary. For further properties of the Lobatto shape functions see Refs. 14–16.

With the obvious separation

$$\mathbf{M}=\tilde{\mathbf{T}}+2(\mathbf{V}-E\mathbf{S}) \tag{16}$$

the matrix elements become independent of the final state energy E. Equation (16) specifies the set of operators for which matrix elements have to be calculated. $\tilde{\mathbf{T}}$ is proportional to the kinetic energy matrix except for the elements involving the surface Lobatto shape function $u_0(r)$; for the latter, an additional surface term has to be included. $\mathbf{V}$ is the matrix representing the (local) potential and $\mathbf{S}$ is the overlap matrix. For a compound basis the following types of matrix elements arise in the LDKL method: (a) matrix elements between two Lobatto shape functions ("free–free"-type); (b) matrix elements between one Lobatto shape function and one GTO ("free-bound"-type); (c) matrix elements between two GTOs ("bound–bound"-type). Application of the Gauss–Lobatto quadrature leads to simple formulas for the evaluation of the "free–free" matrix elements.[14] With an appropriate choice of the volume $\mathscr{V}$ one may evaluate the "bound–bound" matrix elements using conventional procedures.[30] Merely the evaluation of the "free-bound" matrix elements takes some extra effort. In the following this will be discussed for the example of a "free-bound" overlap matrix element between a GTO and a Lobatto shape function.

The present work is based on the linear combination of Gaussian-type orbitals local density functional (LCGTO-LDF) method[31] which draws part of its computational efficiency from the use of spherical harmonic GTOs (Ref. 30) as "bound"-type basis functions,

$$\chi_{l\mu}^{a\alpha}(\mathbf{r})=N^{l,\alpha}C_\mu^l(\mathbf{r}-\mathbf{a})\exp[-\alpha(\mathbf{r}-\mathbf{a})^2]. \tag{17}$$

This type of GTO, here centered at the (nuclear) position $\mathbf{a}$, is a product of a Gaussian with exponent α and a real solid harmonic with quantum numbers (l,μ),

$$C_\mu^l(\mathbf{r})=\left(\frac{4\pi}{2l+1}\right)^{1/2} r^l Y_{l\mu}(\Omega). \tag{18}$$

$N^{l,\alpha}$ is a normalization factor. The "free"-type basis functions $r^{-1}u_i(r)Y_k(\Omega)$ are always defined with respect to the center of the sphere S which is also the origin of the grid of the radial Gauss–Lobatto integration. This structure of the "free"-type basis functions suggests a separation of the overlap integral into an angular and a radial part. For a "free-bound" overlap integral one has

$$S_{\mathrm{fb}}=\left\langle \frac{1}{r}u_i(r)Y_k(\Omega)\middle|\chi_{l\mu}^{a\alpha}(r,\Omega)\right\rangle$$

$$=\omega_i r_i\int d\Omega\, Y_k(\Omega)\chi_{l\mu}^{a\alpha}(r_i,\Omega). \tag{19}$$

Here the radial part has been evaluated using a Gauss–Lobatto quadrature.[29] The "orthogonality" of the Lobatto shape functions with respect to the nodes [see Eq. (15)] implies that for each angle Ω the integration requires simply the evaluation of the GTO at the radial position r_i which is associated with the given Lobatto shape function. This radial integration might seem to be rather approximate, but the sharply peaked Lobatto shape function can be thought of as "sampling" the GTO. Of course, this view is only appropriate for a sufficiently diffuse GTO, i.e., if the exponent α is small enough. On the other hand, a GTO with a very large exponent "samples" the Lobatto shape function. However, these problems may be overcome with an appropriate cutoff technique which excludes Lobatto shape functions peaking in the vicinity of a nucleus from the variational ansatz. The remaining angular part of the integral is evaluated by a single-center expansion of the GTO around the molecular center. The projection onto the spherical harmonics centered at the origin may be carried out analytically.[32] This single-center expansion may be simplified further by separately treating the Gaussian lobe function and the exponent-independent solid harmonic of the GTO. However this implies an additional summation due to the two independent projections. Once the overlap integrals are calculated, the matrix elements involving the kinetic energy operator and the dipole operator may be derived by parameter differentiation and angular momentum coupling with the dipole operator $\mathbf{r}=C_\mu^1(\mathbf{r})$, respectively.[8,30]

Within the LCGTO-LDF implementation of density functional theory, the charge density and the exchange-correlation potential are expanded in a set of auxiliary fitting functions,[31,33]

$$\tilde{\rho}(\mathbf{r})=\sum_j a_j g_j(\mathbf{r}),$$
$$\tilde{v}_{\mathrm{xc}}(\mathbf{r})=\sum_l b_l h_l(\mathbf{r}). \tag{20}$$

The local potential in the effective one-electron Hamiltonian is then given by

$$V=v_n+\tilde{v}_c+\tilde{v}_{\mathrm{xc}}, \tag{21}$$

where v_n and $\tilde{v}_{\mathrm{xc}}$ are the nuclear attraction and the exchange-correlation potential, respectively, and $\tilde{v}_c$ is the classical Coulomb potential, calculated from the fitted charge density,

$$\tilde{v}_c(\mathbf{r})=\int\frac{\tilde{\rho}(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}'=\sum_j a_j\int\frac{g_j(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}'. \tag{22}$$

The expansion coefficients a_j and b_l are obtained from the self-consistent ground state calculation. Because the fitting functions are also Gaussian lobe functions combined with solid spherical harmonics, the formalism for projecting the exchange-correlation potential is the same as that for the projection of the "bound"-type functions onto the "free"-type basis. The Coulomb potential, derived from the fitted charge density according to Eq. (22), involves projections of incomplete gamma functions which are more complicated, but can also be carried out analytically. The projection of the nuclear potential is straightforward.[34] Thus the single-center expansion of the effective local potential at the nodes of the radial Gauss–Lobatto quadrature is obtained. "Bound-free" matrix elements of the potential are evaluated by coupling the two single-center expansions, one of the GTOs of the orbital basis and one of the potential. Additionally, the condition that higher multipole moments of the potential have to vanish near the boundary surface may be checked with the help of the potential expansion.

III. METHODICAL ASPECTS

A. Parameters and basis sets

Any finite volume finite basis variational principle involves several conditions and parameters which affect the quality and the reliability of a calculation. Most of them have been encountered previously, but will be mentioned here again to allow a coherent discussion of all aspects of the present LDKL formalism.

The finite volume we consider is a sphere of radius S with its origin at the center of mass of the molecule. It should be chosen large enough so that all contributions from the "bound"-type basis functions are negligible at its surface. This is required by the boundary conditions inherent to the LDK variational principle and for the application of the Gauss–Lobatto quadrature. On the other hand, too large a radius S increases the number of Lobatto shape functions necessary for a reliable representation of the outgoing waves and thus leads to a larger computational effort. A second criterion for the determination of the radius S is connected to the fact that the wave function outside the finite volume $\mathscr{V}$ is a linear combination of Coulomb waves; thus all higher multipole moments of the potential have to vanish outside $\mathscr{V}$. Experience shows that the criterion concerning the "bound"-type basis functions is sufficiently restrictive to guarantee that the second criterion concerning the multipole moments of the potential is also fulfilled (see Table I).

A pure one-center expansion of a continuum wave function requires much higher angular momenta than those which contribute significantly to the photoionization cross section.[35] In the LDKL approach the additional "bound"-type basis functions supply a rich multicenter basis which allows us to confine the one-center expansion to partial waves which propagate the waves to the volume boundary and thus to infinity. Nevertheless, for each application a convergence study has been carried out to check the convergence of the partial wave expansion of the final state. Exclusion of relevant angular momentum components from the continuum wave function may cause spurious resonances.

As outlined in Sec. II C the calculation of the matrix elements of the potential is based on a multipole expansion of the effective potential. An improper truncation of this expansion leads to an incorrect representation of the Hamiltonian operator. The accuracy of the matrix elements of the potential also has been examined by investigating the convergence of the photoionization cross section with respect to the truncation of the multipole expansion.

TABLE I. Initial state and final state basis sets for the molecules under investigation. Also given are the cutoff interval, the radius S of the sphere and the maximum angular momentum l_p for the multipole expansion of the effective potential. The final state basis set consists of all GTOs of the ground state calculation and of a Lobatto basis of order M (subject to the cutoff), including partial waves up to l_s.

System	GTO ($s/p/d$)	Lobatto M	Lobatto l_s	Cutoff (a.u.)	S (a.u.)	l_p
N_2[a]	13/8/2[d]	60	3	0.5–1.5	25.0	8
CO[b]	13/8/6[e]	60	5	0.5–1.5	25.0	12
C_6H_6[c]	C(10/6/5), H(7/3)[f]	100	7	2.14–5.19	35.0	14

[a]Bond length, 2.068 a.u. (Ref. 37).
[b]Bond length, 2.132 a.u. (Ref. 8).
[c]Bond lengths, 2.6400 a.u. (C–C), 2.0485 a.u. (C–H) (Ref. 68).
[d]Reference 60, polarization functions from Ref. 61.
[e]Reference 60. The d-type polarization functions (Ref. 61) have been suitably augmented (Ref. 62). For details see text.
[f]Derived from $9s/5p$ (C) and $6s$ (H) basis sets from Ref. 60, d-exponents (C) from Ref. 61, and p-exponents (H) from Ref. 69. For details see text.

The number of Lobatto shape function determines the order of the polynomial representation of the continuum wave function. However, because these "free"-type basis functions are associated with the nodes of the Gauss–Lobatto quadrature which is used for carrying out the radial integration of the given matrix elements, the accuracy of the numerical integration is also related to the size of this basis. Furthermore, the quality of the numerical integration implied in the evaluation of the "free-bound" matrix elements depends on the radial position of the nuclear centers. Therefore, for larger molecules, the Lobatto degree M has to be chosen higher than for diatomics; typical values of M range from 60 to about 120 (see Table I).

As shown previously[16] for the system H_2^+, a highly accurate representation of the initial state is required to obtain converged photoionization cross sections. For this system, a very large Gaussian basis set (e.g., $21s/11p/5d$) was found to be necessary in order to achieve an accuracy of better than 1% compared to the "exact" numerical values.[36]

The influence of the ground state "bound"-type basis set was also studied in the case of N_2. The parameters for the "free"-type basis set are as reported in Table I. In all these calculations the final state basis was augmented by the corresponding full ground state GTO basis set. Furthermore no contractions have been applied to the continuum GTOs in order to retain the maximum flexibility of the basis set. In Table II the position of the σ_u shape resonance and its height are compared for different GTO basis sets. Also shown are the height and the position of the

TABLE II. Comparison of the position and the height of the σ_u shape resonance in the photoionization of N_2 from the $3\sigma_g$ orbital with respect to different ground state basis sets. The continuum basis is augmented by the full ground state basis set. For the parameters of the translational basis set see Table I.

Basis set (*s/p/d*)	Total energy (a.u.)	Position of resonance (eV)	Height of resonance (Mb)
10/5/2[a]	−108.6941	29.87	15.74
13/8/2[b]	−108.7015	29.74	14.53
13/8/5[b]	−108.7060	29.74	14.09
16/11/5[b]	−108.7078	29.90	13.99
18/13/5[b]	−108.7075	29.92	13.75
HF–CI[c]		29.30	8.93
Experiment[d]		27.91	9.59

[a]Reference 60, polarization functions from Ref. 61.
[b]Reference 52, polarization functions from Ref. 61. The two original polarization exponents have been augmented by three additional *d*-exponents according to a geometrical series starting from 0.187 65 with a progression of 2.1956.
[c]Reference 37.
[d]Polynomial interpolation of the data from Ref. 38.

resonance from a HF-CI calculation[37] as well as the experimental data.[38] Inspection of Table II reveals that a moderately large basis set like $13s/8p/2d$ leads to sufficiently stable results. The additional *d*-functions introduced in the $13s/8p/5d$ basis set improve the agreement with experiment with respect to the height of the resonance; its position is very close to the experimental value even with the smallest basis set, with deviations of <0.2 eV between the various basis sets. This is significantly less than the total deviation of about 2 eV from the experimental value. The fact that the largest ground state basis set does not produce the lowest total energy is typical for the LCGTO-LDF method.[33] The auxiliary basis sets for the exchange-correlation potential and the charge density are coupled to the orbital basis by an appropriate scaling of the exponents. This results in a different Kohn–Sham Hamiltonian for each basis set. Therefore it may not be expected that a basis set augmentation variationally lowers the total energy.

Using the same set of "bound"-type functions for the continuum wave function calculation as for the ground state calculation has the advantage that only one set of "bound–bound" matrix elements has to be evaluated for the various operators. The resulting large basis set may be nearly linear dependent. A simple way to avoid this problem is to exclude Lobatto shape functions peaking in a volume where a large number of GTO functions is located, i.e., close to the nuclear centers. The exclusion of Lobatto shape functions from spherical shells near the nuclei also helps to avoid numerical problems that may arise from the Gauss–Lobatto quadrature of "free-bound" matrix elements involving GTOs with large exponents. In the case of N_2, these shells are chosen to span a radial interval from 0.5 to 1.5 a.u. The sensitivity of the results with respect to a variation of this radial cutoff interval was found to be small; this demonstrates both the completeness of the compound basis set and the quality of the underlying numerical integration. In addition to the foregoing procedure, linear dependencies were suppressed by constructing an orthogonal basis set by diagonalizing the overlap matrix of the full compound basis and by eliminating eigenvectors associated with small eigenvalues (typically $<10^{-5}$). Variation of this threshold from 10^{-4} to 10^{-6} led only to very small changes in the cross section.

B. The effective potential for the photoelectron

For the example of H_2^+ the LDKL formalism has been shown to yield an accurate representation of molecular continuum wave functions.[16] Therefore, the variational principle and the numerical approximations implied within this method may be regarded as sufficiently stable and accurate. However, in the case of many-electron systems the potential representing the interaction of the photoelectron with the remaining ion is no longer given exactly, but has to be approximated by a suitable one-electron potential.

In the LDKL formalism one is free to choose a final state potential which is different from that determining the initial state. In the following, different types of final state potentials will be discussed and their influence on the σ_u shape resonance observed in the ionization of the $3\sigma_g$ orbital of N_2 will be considered. This shape resonance may be thought of as being produced by resonant trapping of the *f* partial wave within the potential well.[38] Another interpretation of shape resonances to which we will refer later on relates them to unoccupied orbitals.[39–41] This picture allows a prediction of the approximate energy and the symmetry of the resonances by identifying the virtual valence orbitals from minimal basis set calculations with resonant states.

A crucial point in the choice of the final state potential is the treatment of the so-called self-interaction, the unphysical Coulomb interaction of an electron with its own charge distribution. The standard nonlocal N-electron potential appearing in the Hartree–Fock Hamiltonian operator is self-interaction free with respect to the occupied orbitals corresponding to the N bound electrons of the ground state of the system under consideration. For unoccupied orbitals which are of interest for the description of photoelectrons this does not hold. The N-electron Hartree–Fock potential as the final state potential has the wrong asymptotic behavior, namely that of a neutral system and not that of a positive ion. Therefore, in Hartree–Fock based methods describing photoelectrons, a $N-1$-electron potential is used which is constructed by removing the contributions of the initial state orbital from the standard N-electron potential. The resulting final state potential then has the correct asymptotic behavior. Because in the $N-1$-electron potential the contributions from the $N-1$ orbitals, which are not ionized, are considered to be unaffected by the ionization process, this construction of the final state potential implies the so-called frozen core approximation.

In a density functional method for photoelectrons a natural choice for the final state potential is the self-consistent N-electron ground state potential V_N. The exact local effective Kohn–Sham potential, acting on an electron moving in the field of the other electrons of a multielectron system, is self-interaction free. In contrast to the

N-electron Hartree–Fock potential, the exact N-electron Kohn–Sham potential exhibits the correct behavior for large distances from the center of a (neutral) molecule,[42] i.e., it is asymptotically identical to the Coulomb potential of a positive ion. (Note that the Kohn–Sham or effective potential V_N of this work is the sum of the external nuclear potential plus that potential denoted in Ref. 42 as the effective potential.) If the potential V_N is separated into the nuclear potential v_n, the classical electron–electron repulsion potential v_c, and the exchange-correlation potential v_{xc} [see Eq. (21)] then it follows from the known asymptotic behavior of v_n, v_c, and V_N (Ref. 42) that v_{xc} asymptotically has to be identical to the potential of a positive point charge. Finally note that by choosing the N-electron Kohn–Sham potential as final state potential also the frozen core approximation is made.

In actual applications of density functional theory several approximations have to be made. In the method introduced here these are as follows:

(a) The exact Kohn–Sham potential is unknown. Therefore well established potentials according to the local density approximation (LDA) (Refs. 43,44) are used widely which for bound systems lead to very good results. One purpose of the present work is to investigate whether this is true also for unbound systems. The findings displayed later on give evidence for an affirmative answer to this question. However, in the LDA the exchange-correlation potential v_{xc} is approximately proportional to the electron density to the power of 1/3 and therefore, like this quantity, falls off exponentially. Therefore the potential v_{xc} does not have the correct asymptotic but is effectively zero at large distances. Because the potentials v_n and v_c cancel each other for large distances, the effective potential V_N corresponds asymptotically to the potential of a neutral system instead of a positive ion.

Furthermore, in this work the exchange-correlation potential is represented by Gaussian basis functions leading to the potential $\tilde{v}_{xc}$ of Eq. (20). Gaussian functions, however, decay exponentially with the square of the distance from their center (usually the atoms). Therefore for large distances the exchange-correlation potential $\tilde{v}_{xc}$ dies off even faster than the actual LDA exchange-correlation potential. It is important to note that this wrong asymptotic behavior is not a consequence of the choice of the N-electron Kohn–Sham potential as final state potential, but of the LDA and of the representation of the resulting exchange-correlation potential by a Gaussian basis set. An evident improvement of this approximate final state potential is a Latter tail correction or an alternative correction introduced below. Then, the local effective potential is represented according to Eqs. (20) and (21) in the inner region where it is determined by the individual system, whereas in the outer region where it is independent of a specific system it is set equal to the Coulomb potential of a positive ion, i.e., to its correct asymptotic form.

(b) Except for the highest occupied orbital the Kohn–Sham orbital energies do not represent the corresponding ionization potentials.[45] As a consequence, experimental ionization energies are used here. As mentioned in Sec. II A, alternatively the ionization energies could have been determined by ΔSCF calculations. Furthermore, note that the Kohn–Sham approach formally is not aimed at a description of excited states. However, as the results of this work demonstrate, one-electron states generated with a method based on density functional theory provide a reasonably good approximate description of photoelectrons.

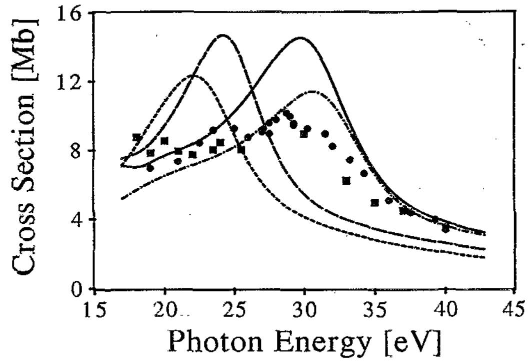

FIG. 1. Influence of various choices of the final state potentials exemplified by the ionization from the $3\sigma_g$ orbital of N_2. N-electron ground state potential (solid line), $N-1$-electron potential (short dashes), transition state potential (long dashes), Latter tail corrected potential (dot dash). For the parameters of the calculations see Table I.

The foregoing discussion has focused on fundamental aspects concerning the final state potential and its asymptotic behavior. However, for the calculation of total photoionization cross sections which is the main focus of the present work, the asymptotic behavior of the potential is not of dominant influence. This is demonstrated by the general success of ST methods.[4–6,8] Indeed, it has been shown that cross sections calculated from the N-electron LDF potential without Latter tail correction are in good agreement with experiment.[8] Also it turned out that the N-electron potential yields satisfactory resonance energies for valence ionization processes. For the method introduced here, Fig. 1 shows that the resonance position calculated using V_N is in good agreement with experiment and with other calculations. However, Fig. 1 also shows that the resonance height is rather overestimated. This may be attributed to deficiencies in the potential approximation discussed above and to the neglect of polarization effects due to the ionizing dipole field. The latter may be addressed by treating the photoabsorption within time-dependent density functional theory (TDDFT).[46,47] The TDDFT formalism is aimed at the calculation of the (complex) frequency-dependent polarizability whose imaginary part is directly related to the photoionization cross section. The TDDFT method is capable of including the response of the "inactive" electrons to the external field, i.e., the electrons which are not ionized. This generally improves the agreement with experiment, but significantly increases the computational effort because the usual dipole operator r is replaced by a spatially varying and frequency-dependent local field[47] which has to be recalculated for each photon energy under investigation. Thus the TDDFT so far only has been applied to atoms[48] and small molecules.[49]

As mentioned above a common method for adjusting the asymptotic behavior of the N-electron potential is the Latter tail correction[50] which was first applied in atomic calculations. It consists of a simple replacement of the approximate effective potential by the asymptotically correct ionic potential outside that radius where the latter is more attractive than the atomic potential V_N. This procedure is also used in the CMS method.[3] There the correction can be applied directly because of the spherical averaging of the potential outside the muffin-tin sphere which encloses the molecule. Since the multipole components of the local effective potential are calculated explicitly in the LDKL method, it is rather straightforward to apply a Latter tail correction to the monopole component of the effective potential from that radius on where it exceeds the asymptotically correct ionic potential. Only for the matrix elements between "bound"-type basis functions a numerical correction becomes necessary. However, the applicability of this technique is restricted to molecules with a fast converging multipole representation of the potential. At the radial distance from where on the Latter tail correction is employed the higher multipole components of the potential should vanish. Otherwise the angular structure of the potential will be modified. Figure 1 shows that the position of the resonance is slightly shifted to higher energies, but the quantitative agreement with the experimental cross section is improved significantly with a Latter tail corrected potential.

In cases in which the Latter tail correction is inappropriate because the effective potential contains significant contributions from higher multipole components outside the Latter radius an alternative asymptotic correction may be applied. In a similar spirit one corrects the monopole component of the quantity $r \cdot V_{\text{eff}}(r)$ from that radius on where all higher multipole components have decayed below a given threshold by switching gradually to the correct asymptotic form outside the finite volume. Test calculations on the photoionization cross section of the valence orbitals of benzene using a completely unmodified self-consistent N-electron ground state potential and spherical Bessel functions and Neumann functions for the asymptotic form of the wave function yielded practically identical cross sections as those which were obtained with the asymptotically corrected N-electron potential and the corresponding wave functions. This demonstrates that the photoionization cross sections in the case of benzene are quite insensitive to the asymptotic behavior of the used effective potential. However, for formal reasons the corrected potential is preferable as the photoelectron wave functions then are asymptotically of the correct form. The present multipole dependent asymptotic correction leaves also the asymmetry parameter essentially unchanged.[51]

Alternatively one might construct a potential in analogy to the nonlocal $N-1$-electron potential used in Hartree–Fock related methods by reducing the occupation of the initial state molecular orbital from which the electron is emitted. Then one recalculates the potential expansion (20) keeping fixed the charge distribution of the $N-1$ remaining occupied orbitals. As a Kohn–Sham potential this $N-1$-electron potential is related to an $N-1$-electron system and therefore the corresponding photoelectron states describe the ionization of a positive ion. On the other hand, the $N-1$-electron potential is not the self-consistent ground state Kohn–Sham potential of the corresponding ion. Therefore this $N-1$-electron potential is physically not well defined. Nevertheless, due to the local density approximation used, a cancellation of errors causes this $N-1$-electron potential to exhibit the correct asymptotic Coulomb behavior. However, this is accidental and should not be used as an argument in favor of the $N-1$-electron potential. As expected, the σ_u resonance of N_2 is dramatically shifted to lower energies and its height is reduced by about 2 Mb (see Fig. 1).

An attempt to choose a potential that takes into account part of the relaxation of the $N-1$ electrons (which are not ionized) is the transition state technique which is often used in density functional theory for the calculation of ionization energies[2] and photoionization cross sections.[3] In this scheme only half an electron is removed from the orbital under consideration with a subsequent self-consistent determination of all orbitals. However, as seen in Fig. 1, this technique affects the results in an unfavorable way similar to the $N-1$-electron potential. The objections concerning the asymptotic behavior as discussed for the $N-1$-electron potential are in a similar way also valid for the transition state potential.

From this comparison we conclude that in the case of a rapidly converging multipole expansion of the effective potential the Latter tail correction leads to the best results. For N_2, CO and other small molecules this type of adjusted potential will be used. If the shape of the potential deviates strongly from spherical symmetry the use of the N-electron potential in combination with the introduced multipole dependent correction is preferable, especially since it has been demonstrated to yield good agreement for resonance energies which are of primary interest for the interpretation of experimental results. Thus the photoionization of the valence orbitals of benzene which will be discussed in Sec. IV B was investigated using the N-electron potential with a multipole-based asymptotic correction.

IV. APPLICATIONS

A. Application to the photoionization of CO

To further test the LDKL method we have applied it to CO. This molecule has been studied quite intensively by experiment[38,53,54] and by a variety of different theoretical methods.[6,8,55–59] Thus, the obtained spectra have been already discussed in full detail elsewhere. The aim here is it to compare our results to existing data in order to illustrate the accuracy and the computational effort of the present method.

The calculations were performed using $13s/8p/6d$ basis sets for both C and O obtained from standard $13s/8p$ basis sets for the two atoms.[60] The six d-polarization expo-

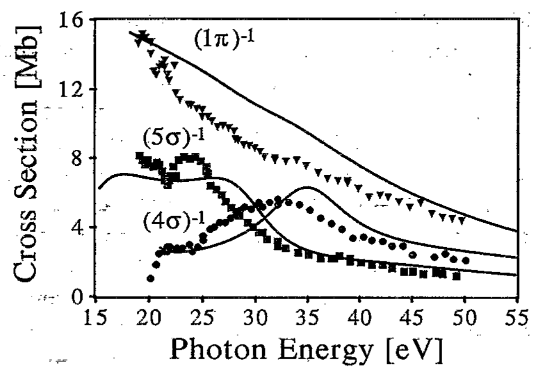

FIG. 2. Calculated and experimental cross sections for the photoionization from the 4σ (circles), the 5σ (squares), and the 1π (triangles) valence orbitals of CO. A Latter tail correction was used. The parameters of the calculation are given in Table I.

nents (C: 0.0965, 0.2413, 0.5416, 0.6000, 1.2189, 3.7697; O: 0.2413, 0.5416, 1.1500, 1.2189, 3.7697, 9.4243) have been obtained by suitably scaling standard d-polarization exponents.[61,62] For all of the investigated initial orbitals, 4σ, 5σ, and 1π, the same final state potential has been applied; the N-electron potential with an exchange-correlation contribution according to the Vosko–Wilk–Nusair parametrization of the local density approximation[63] and a Latter tail corrected potential ensuring the proper asymptotic behavior.

The results for the selected initial orbitals are given in Fig. 2. It can be seen that the reported cross sections are in excellent quantitative agreement with the experimental results. The σ^* shape resonance is shifted to higher energy by about 2 eV; it exhibits a sharper peak than the experimental cross section. This can be rationalized by the fact that experiments generally yield line shapes of resonances which are broadened due to vibrationally averaging over different nuclear configurations.[64–66] The σ^* shape resonance is known to be very sensitive to changes of the internuclear separation.[64] If we compare the LDK results for the $(4\sigma)^{-1}$ cross section with those from ST imaging[8] we can see that the latter method finds the resonance position also shifted to slightly higher energy with a shape closer to the experimental one. This is a consequence of the smoothing inherent to ST imaging and not of physical origin. The structure below 21 eV in the $(5\sigma)^{-1}$ cross section has been attributed to autoionization[8] and therefore does not appear in the present description.

The parameters of the LDKL calculation are specified in Table I. Although the resonance is predominantly of f wave character, we found that an expansion of the continuum wave function using angular momenta up to $l_s=3$ leads to spurious resonances at higher energies. This is in contrast to the case of N_2 where the inclusion of higher partial waves does not yield any further improvements.

B. Application to the photoionization of benzene

The photoionization of the valence orbitals of benzene has been investigated previously both experimentally[67] and theoretically.[8,67] Yet in various cases the calculated and the experimental data differ substantially. While CO and N_2 have been studied theoretically with a variety of methods, the only theoretical methods so far applied to benzene are the ST imaging approach[8] and the CMS method.[67] Because of its plane geometry, benzene is a molecule for which the muffin-tin partitioning necessary for the CMS method implies a rather drastic simplification. Therefore, this molecule may serve as an example for the advantages of a method which is able to treat the molecular potential without shape approximation.

We have investigated the ionization from the four highest occupied MOs of benzene, the $3e_{1u}$, $1a_{2u}$, $3e_{2g}$, and $1e_{1g}$ orbitals. The geometry was chosen as in Ref. 68. As discussed above, the nonspherical shape of the molecule causes a slowly converging multipole expansion of the potential. Thus the introduced multipole dependent asymptotic correction of the N-electron potential was applied. The starting radius of the multipole dependent correction is chosen at 10.9 a.u. At this point the higher multipole components of the potential are smaller than 10^{-2} a.u. As mentioned above, essentially identical results have been obtained with the unmodified self-consistent N-electron ground state potential which also leads to satisfactory results in the ST imaging treatment.[8]

The ground state calculation was performed using a $7s/3p$ basis set for H and a $10s/6p/5d$ basis set for C. Both basis sets were obtained from standard basis sets[60] by adding an additional diffuse exponent for each angular momentum (continuing the given exponents according to a geometrical series). To support higher angular momenta on the nuclear centers, additional polarization functions [H: p-type (Ref. 69), C: d-type (Ref. 61)] were included. In doing so the p-set for hydrogen was enlarged by one diffuse exponent forming a geometrical series with the two original exponents. The five d-exponents (C) were also constructed as a geometrical series (starting from 0.133 767 with a progression of 1.7045), based on the two standard polarization exponents.[61] Because of the diffuse exponents, a sphere radius S of 35 a.u. was chosen.

In order to probe the convergence of the multipole representation of the potential, a series of calculations with increasing angular momentum l_p for the potential expansion, but with a fixed reduced Lobatto order of $M=20$ and a partial wave expansion up to $l_s=7$ was carried out. The reduction of the Lobatto degree significantly lowers the computational effort required for the test calculation, but nevertheless provides reliable information on the convergence of the multipole expansion. For the angular momentum of $l_p=14$ convergence was reached; a potential expansion with $l_p=16$ did not lead to any changes in the results. After choosing a Lobatto order of $M=100$ in accordance with the more extended finite volume, partial wave expansions of the continuum wave function with $l_s=5$ and $l_s=7$ were compared. In the case of photoionization from the $1e_{1g}$ orbital, the cross section for $l_s=5$ showed spurious structures at about 35 eV which disappeared for the increased angular momentum $l_s=7$ and which therefore were identified as artifacts caused by an insufficient partial

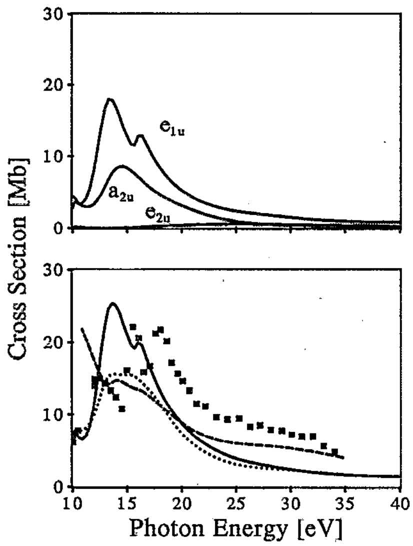

FIG. 3. Cross section for the photoionization from the $1e_{1g}$ orbital of benzene. Upper panel, partial cross sections for final states of different symmetries. Lower panel, total photoionization cross section; experimental data (squares) (Ref. 67), results of a LDKL calculation (solid line, this work), of a ST imaging calculation (dots) (Ref. 8), and of a CMS calculation (dashes) (Ref. 67).

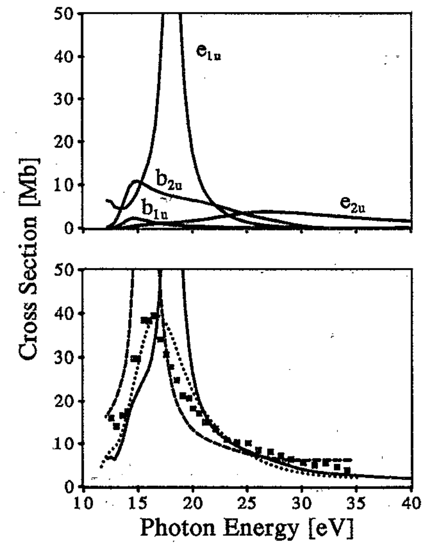

FIG. 4. Cross section for the photoionization from the $3e_{2g}$ orbital of benzene. Upper panel, partial cross sections for final states of different symmetries. Lower panel, total photoionization cross section; experimental data (squares) (Ref. 67), results of a LDKL calculation (solid line, this work), of a ST imaging calculation (dots) (Ref. 8), and of a CMS calculation (dashes) (Ref. 67).

wave expansion. All further calculations were carried out by expanding the continuum wave function into partial waves of up to $l_s=7$. Of course, only a subset of these partial waves contribute to the various photoionization processes, depending on the dipole selection rules for the point group symmetry D_{6h} and on the investigated initial molecular orbital. All pertinent parameters of the calculation are collected in Table I.

Because benzene possesses a manifold of low-lying unoccupied orbitals the calculated spectra will show a variety of shape resonances. As discussed above, these may be correlated with the virtual orbitals as obtained from a minimal basis SCF calculation.

1. Ionization from the $1e_{1g}$ orbital

In Fig. 3 we present the results for photoionization from the $1e_{1g}$ orbital, the highest occupied molecular orbital of benzene. It can be seen that the CMS calculation[67] completely fails to reproduce the experimental data whereas the calculation based on the ST imaging formalism[8] correctly reflects the resonant structure near the photon energy of 15 eV. The LDKL results show that the resonant structure is actually built from two neighboring shape resonances of e_{1u} final state symmetry which cannot be resolved within the ST imaging calculation. The corresponding virtual orbitals are $4e_{1u}$ and $5e_{1u}$. However, the peak found in the a_{2u} final state channel cannot be explained within the LCAO-MO scheme. Rather it has to be associated with the resonant trapping of the corresponding f wave, as confirmed by a phase shifts analysis of the asymptotic partial waves. As in the ST imaging approach, the a_{1u} final state symmetry does not contribute. The LDKL results exhibit the correct behavior at threshold. Inspection of Fig. 3 reveals the similarities between the results from the ST technique and from the present method; the ST cross section in general shows the same qualitative behavior and produces resonances at about the same photon energies, but fails to resolve closely adjacent features. Of course, an eigenphase analysis as used in the analysis of the a_{2u} shape resonance is only possible if the asymptotic form of the continuum wave function is available as for the present method.

2. Ionization from the $3e_{2g}$ orbital

The photoionization cross sections of the $3e_{2g}$ orbital are very similar for all three methods; the shape resonance in the e_{1u} continuum dominates all other channels. From Fig. 4 it can be seen that our calculation produces the resonance shifted toward higher energies by about 2 eV. Additionally as in the CMS calculation, the resonance height is largely overestimated. The ST imaging results benefit from their inherent lower resolution which broadens the sharp resonance and thus leads to a good, but fortuitous agreement with the experimental data, as proven by the LDKL results for the same potential. In this case the inclusion of local shielding effects, as taken into account by the TDDFT,[46,47] is expected to improve the agreement with the experimental data.

3. Ionization from the $1a_{2u}$ orbital

Via the dipole operator the $1a_{2u}$ level couples only to final states of a_{1g} and e_{1g} symmetry. Immediately above the threshold, antibonding σ*C–H contributions generate a virtual valence orbital ($4a_{1g}$). This is reflected in the rise of both, experimental and calculated cross sections. Figure 5 reveals that the CMS method is unable to yield the correct results in this case. Only the LDKL results are in reason-

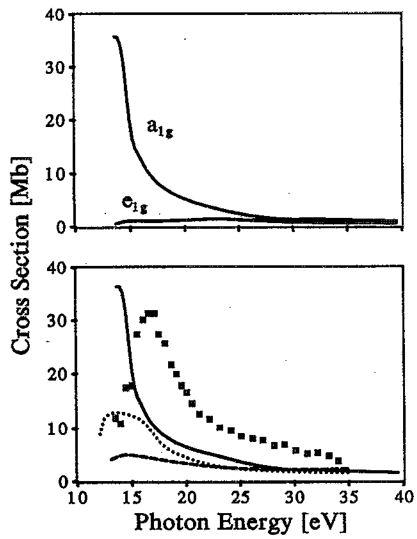

FIG. 5. Cross section for the photoionization from the $1a_{2u}$ orbital of benzene. Upper panel, partial cross sections for final states of different symmetries. Lower panel, total photoionization cross section; experimental data (squares) (Ref. 67), results of a LDKL calculation (solid line, this work), of a ST imaging calculation (dots) (Ref. 8), and of a CMS calculation (dashes) (Ref. 67).

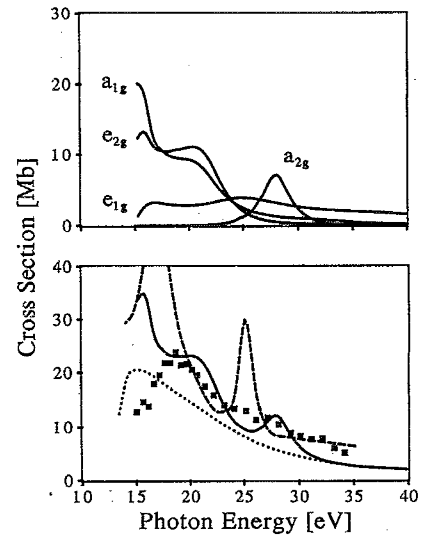

FIG. 6. Cross section for the photoionization from the $3e_{1u}$ orbital of benzene. Upper panel, partial cross sections for final states of different symmetries. Lower panel, total photoionization cross section; experimental data (squares) (Ref. 67), results of a LDKL calculation (solid line, this work), of a ST imaging calculation (dots) (Ref. 8), and of a CMS calculation (dashes) (Ref. 67).

able agreement with experiment. However for a proper evaluation, experimental uncertainties have to be taken into account. The cross section for this initial state may be too high due to normalization problems caused by vibrational broadening that leads to an overlap of this band with the more intense $3e_{2g}$ band.[67]

4. Ionization from the $3e_{1u}$ orbital

Results for the photoionization from the $3e_{1u}$ orbital are shown in Fig. 6. Both ST imaging and the LDKL method yield reasonable agreement with the experimental data.[67] The results from the CMS method suffer from a drastic overestimation of two shape resonant processes. As can be seen from an analysis of the partial cross sections, the shape resonance in the a_{2g} channel is also present in the ST imaging and the LDKL calculation, but much less pronounced. Above 18 eV photon energy the LDKL results reproduce the experimental data satisfactorily, but overestimate the experimental values near the threshold.

V. CONCLUSIONS

We have shown that an accurate and reliable representation of electronic continuum wave functions can be obtained via the logarithmic derivative Kohn (LDK) variational principle with a combined basis set containing "bound"-type basis functions (GTOs) and, as translational basis, Lobatto shape functions. The Lobatto shape functions are ideally adapted to the boundary conditions of the variational principle and, in addition, allow a very efficient integral evaluation. Especially in connection with a local effective one-electron potential from density functional theory, the present method furnishes a useful tool for the description of the photoelectron dynamics of multielectron systems, but requires only rather moderate computational efforts. This demonstrates that density functional theory although formally aimed on ground state properties can contribute to the investigation of ionization processes.

The LDKL approach allows for a free choice of the final state potential. It was found that an asymptotically corrected (Latter tail) N-electron Kohn–Sham potential yields good agreement with the experimental data for small molecules whose potential is of almost spherical symmetry. For larger asymmetric molecules, an asymptotic correction based on the decay of higher multipole components of the N-electron Kohn–Sham ground state potential is preferable. It results in slightly overestimated resonance heights, but accurate resonance positions. The influence of the various possible choices of the effective photoelectron potential have been demonstrated for the example of N_2. For this molecule it also has been shown that the calculated cross sections are relatively insensitive to the initial state representation. Calculations of the partial cross sections for ionization from the valence orbitals of CO illustrated the accuracy of the LDKL method.

An important advantage of the LDKL method is its applicability to comparatively large systems which in the past have been reserved to the Stieltjes–Tchebychev (ST) imaging or the continuum multiple scattering (CMS) methods. For the example of benzene we have demonstrated that the LDKL method does not suffer from the well-known deficiencies of the ST and the CMS method. Particularly the results for the ionization of the highest occupied molecular orbital of benzene, $1e_{1g}$, were clearly superior to those obtained via the CMS method. Although the ST imaging method also gives satisfactory results, it precludes a complete analysis of the results because the continuum wave function is not available explicitly. In contrast to the ST formalism, the LDKL method allows for an eigenphase analysis and the analysis of the angular distribution of the photoelectrons (represented by the β parameter). Furthermore the LDKL formalism is clearly superior to the ST imaging method when narrow spectral features have to be resolved.

We believe that the LDKL method will serve as a useful tool for the interpretation of photoelectron spectra of moderately large molecules. Indeed it has been applied successfully to the metal carbon-cage complex $Ce@C_{28}$.[70,71] The LDKL method may also be used for the interpretation of core level spectra as will be shown in forthcoming publications.

ACKNOWLEDGMENTS

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APPENDIX: COMPARISON OF THE LDK METHOD WITH THE FVV METHOD

In the following we will comment on the relationship between the FVV (Ref. 12) and the LDK method. Both formalisms apply a rather similar variational principle, but the FVV method introduces some arbitrariness by circumventing the explicit integration of the surface derivative (1). The comparison of the two methods becomes particularly transparent if the finite volume $\mathscr{V}$ is chosen to be a sphere and if the underlying basis set in both cases is chosen to obey the LDK boundary conditions,[16]

$$u_i(S)=\delta_{i0}, \quad (i=0,\ldots,M), \tag{A1}$$

i.e., only one function of each angular momentum contributes at the surface $\partial\mathscr{V}$. The main difference between the FVV and the LDK formalism results from the treatment of the surface integral $\langle\psi|\partial_n\psi\rangle_{\partial\mathscr{V}}$ which is a consequence of the integration of the kinetic energy term over a finite volume. The FVV method starts from the assertion

$$\langle\psi|\partial_n\psi\rangle_{\partial\mathscr{V}}=\gamma\langle\psi|\psi\rangle_{\partial\mathscr{V}}. \tag{A2}$$

This condition introduces an additional degree of freedom (the proportionality constant γ) into the formalism and replaces the explicit calculation of the surface derivative integral by a simple surface overlap integration. In the LDK method this surface derivative integral also occurs, but does not affect the solution because of a skillfully chosen boundary condition[16] which even reduces the degrees of freedom for the trial function ψ.

The stationary condition for the functional $\mathscr{L}$ (1) yields for the FVV method the following generalized eigenvalue problem:[12]

$$\mathbf{MD}=\mathbf{\Delta D\Gamma}. \tag{A3}$$

Here, $\mathbf{M}$ consists of the matrix elements of the energy dependent functional (12), $\mathbf{\Delta}$ represents the surface overlap integrals

$$\Delta_{ik,i'k'}=\langle f_{ik}|f_{i'k'}\rangle_{\partial\mathscr{V}} \text{ with } f_{ik}(r,\Omega)=\frac{1}{r}u_i(r)Y_k(\Omega), \tag{A4}$$

$\mathbf{D}$ are the resulting eigenvectors, and $\mathbf{\Gamma}$ is a diagonal matrix containing the eigenvalues γ_i which are the proportionality factors defined in Eq. (A2). Assuming an ordering of the basis functions where the "surface" functions are placed first [see Eq. (11)], the matrix $\mathbf{\Delta}$ simplifies to

$$\mathbf{\Delta}=\begin{pmatrix}\mathbf{E} & \mathbf{0}\\ \mathbf{0} & \mathbf{0}\end{pmatrix}, \tag{A5}$$

where $\mathbf{E}$ is the unit matrix. This is a consequence of condition (A1). Introducing the analogous partitioning for the matrices $\mathbf{M}$ and $\mathbf{D}$, the eigenvalue problem may be rewritten,

$$\begin{pmatrix}\mathbf{M}_{00} & \mathbf{M}_{01}\\ \mathbf{M}_{10} & \mathbf{M}_{11}\end{pmatrix}\begin{pmatrix}\mathbf{D}_0\\ \mathbf{D}_1\end{pmatrix}=\begin{pmatrix}\mathbf{D}_0\mathbf{\Gamma}\\ 0\end{pmatrix}. \tag{A6}$$

If the solutions are linearly independent on the surface $\partial\mathscr{V}$, the matrix $\mathbf{D}_0$ may be inverted. Then a comparison with Eqs. (13) and (14) leads to the identification

$$\begin{aligned}\mathbf{C}&=\mathbf{D}_1\mathbf{D}_0^{-1},\\ \mathbf{L}&=\mathbf{D}_0\mathbf{\Gamma}\mathbf{D}_0^{-1}.\end{aligned} \tag{A7}$$

Thus we have shown that both variational principles lead to the same results if the basis functions are chosen to obey the LDK boundary condition (A1). In this case, the proportionality constants γ_i (A2) are the eigenvalues of the matrix of logarithmic derivatives $\mathbf{L}$ (1).

The main differences between the FVV and the LDK procedures may be summarized as follows. The LDK method poses clear surface boundary conditions on the trial functions, reflected in the properties of the underlying basis set. On the other hand, the FVV method in its original form[12] allows for an entirely free choice of the basis set; all basis functions may contribute on the surface. This results in a genuinely more complicated structure of the occurring surface integrals. However, this difficulty is circumvented in the FVV method by replacing the surface derivative integral by the simpler surface overlap integral at the expense of introducing an additional degree of freedom into the formalism. Thus, instead of a system of linear equations as in the LDK formalism, a generalized eigenvalue problem has to be solved in the FVV method. Furthermore in the original formulation,[12] the FVV method implies a manifold of open channels as determined by the total number of basis functions. This manifold has to be reduced appropriately in order to achieve a set of genuine solutions which are linearly independent on the surface $\partial\mathscr{V}$. This is in contrast to the LDK method where one offers a well-defined set of physically relevant open channels, the convergence of which may be tested in a systematic and straightforward fashion.

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